

Trainings ▾

Preparatory Steps

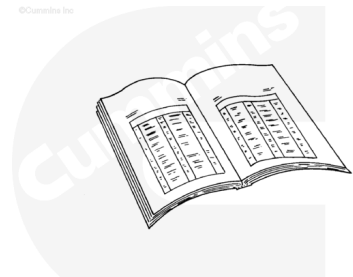
⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain engine oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Remove the oil transfer or suction tube. Refer to Procedure 007-040 (</qs3/pubsys2/xml/en/procedures/101/101-007-040-tr.html>).



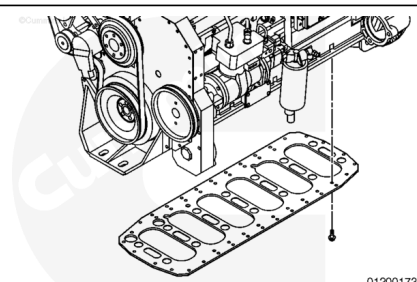
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Remove

Remove and discard the 18 Torx™ head capscrews.

Remove the block stiffener plate from the chassis.

Torx™ capscrews require a size 45 Torx™ socket.

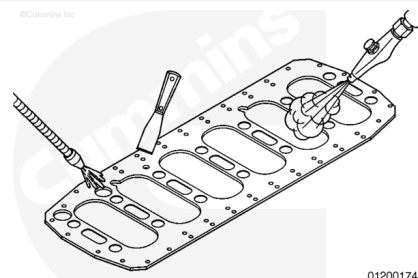


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Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove all sealant material and oil from the cylinder block mating surface.

Thoroughly clean the block threaded capscrew holes.

Remove all gasket and sealant material from the stiffener plate.

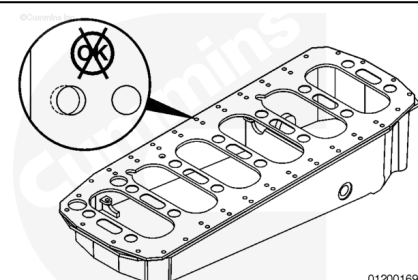
Thoroughly clean all oil from both surfaces of the stiffener plate with solvent.

Dry with compressed air.

Inspect the stiffener plate for cracks or other damage.

Take note of the following:

- The machined-out areas are for main cap and capscrew clearance.
- Oil pump sump suction tube orientation.

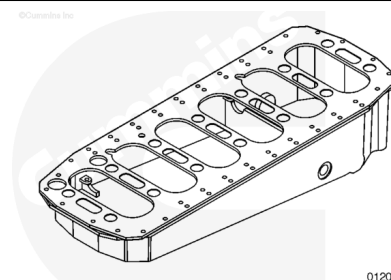


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Machine

⚠ WARNING ⚠

To reduce the possibility of personal injury, keep hands, long hair, jewelry, and loose fitting or torn clothing away from fans and other moving parts.



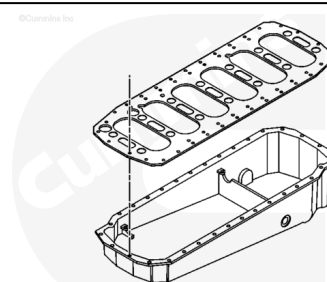
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Note : This machine step applies **only** to cast aluminum oil pan modification.

Non-EGR Aluminum Oil Pan Modification

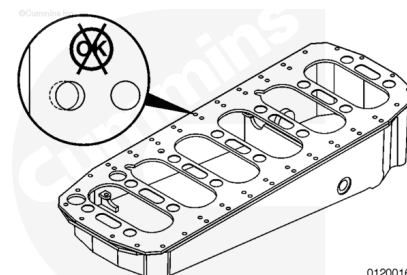
These modifications to the cast aluminum oil pan **must** be made prior to initial stiffener plate installation.

Orient the block stiffener plate onto the oil pan to the correct upfit position on the oil pan mounting flange.



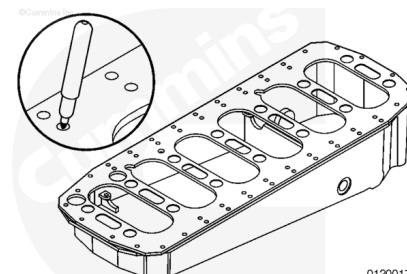
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Align the stiffener plate capscrew holes to the oil pan mounting flange holes.



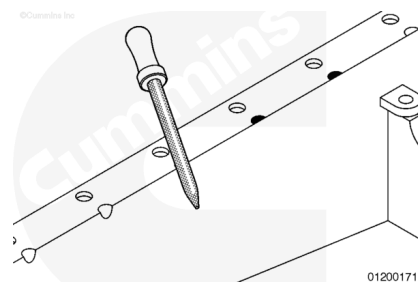
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Locate and mark the 18 Torx™ head screw holes on the oil pan mounting flange.

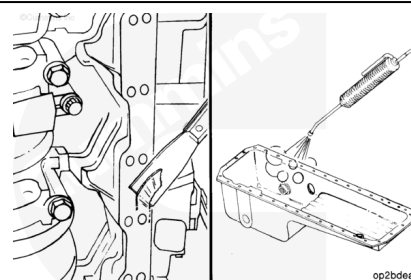


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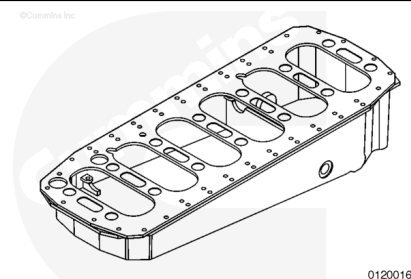
Use a round file to carefully remove enough material in the oil pan mounting flange to accommodate the 2-mm Torx™ head screw protrusion.



After completing all 18 notches, thoroughly clean the pan, removing all aluminum debris.



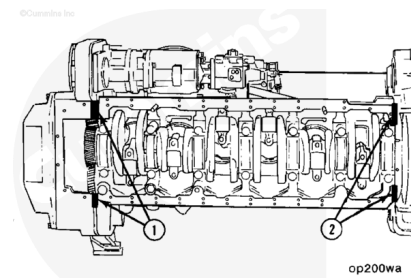
Match the oil pan mounting surface to the stiffener plate and inspect for proper fit.



Install

⚠ CAUTION ⚠

Too much sealant applied to the stiffener plate can drip into the lubrication pan assembly. This excess sealant can become lodged in the oil suction tube screen and cause engine damage.

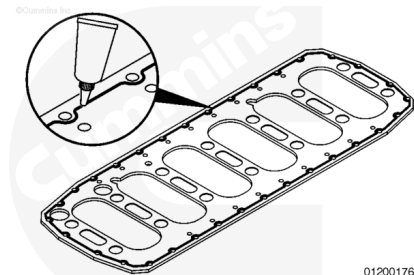


Install four evenly spaced (1 and 2) 10 cm [4 in] long guide pins in the block oil pan capscrew holes.

These guide pins aid in the stiffener plate and oil pan assembly.

Apply a 2 mm [0.0625 in] bead of sealant on the block contact area of the stiffener plate along the oil pan capscrew holes.

Note : The recessed areas in the stiffener plate allow main cap and capscrew clearance. This is the side that faces the block.

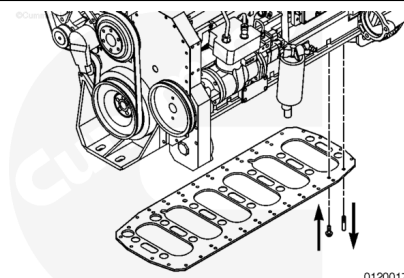


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Maneuver the stiffener plate into place under the engine in the chassis.

Use the guide pins to align the stiffener plate into place and install new Torx™ head capscrews.

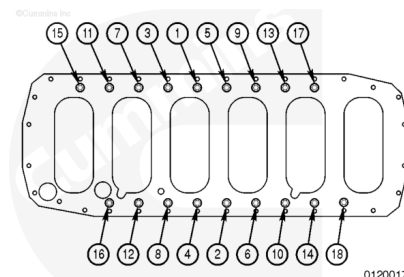
The guide pins can be removed, or left in place to assist in oil pan installation.



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Tighten the Torx™ head capscrews in the sequence shown.

Torque Value: 47 n•m [35 ft-lb]



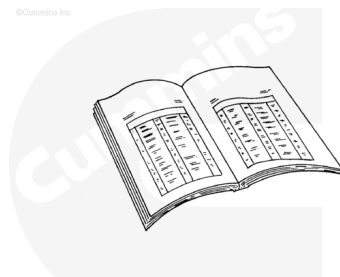
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Finishing Steps

⚠ CAUTION ⚠

Make sure the oil transfer tube designed for the block stiffener plate is used. Failure to install the correct oil transfer tube will result in engine damage.

- Replace the discarded oil transfer tubes with the correct oil transfer tubes.
- Install the oil transfer tubes. Refer to Procedure 007-040 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-040-tr.html>)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)



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- Fill the engine with oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Operate the engine and check for leaks.

Last Modified: 23-Jan-2009
